UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

VECTOR CALCULUS IV: The Stokes theorem

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0.1 Introduction

This chapter is on the Stokes theorem which together with Green theorem and the divergence or Gauss theorem (to be studied in the next chapter) play important roles in establishing fundamental results in many branches of Science and Engineering. As we will see from their statements, the three theorems share a common principle which is easy to identify.

0.2 The Stokes theorem

0.2.1 Curl and Divergence

Let U be a region in $\mathbb{R}^3$ and let $\vec{F} : U \to \mathbb{R}^3$ a vector field with

$$\vec{F}(x,y,z) = (P(x,y,z), Q(x,y,z), R(x,y,z)) \quad \forall (x,y,z) \in U ,$$

with the functions P, Q, R having continuous partial derivatives. Then,

$$\boxed{\mathrm{Curl}(\vec{F}) = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)} . \tag{1}$$

The notion of curl plays an important role in Physics. For instance, in fluid mechanics, where $\mathrm{curl}(\vec{F})(x,y,z)$ is interpreted as a measure of the rotation of particles of the fluids in small area surrounding the point (x,y,z). Another way of illustrating what $\mathrm{curl}(\vec{F})$ means is to consider for instance a flowing fluid with $\vec{F}(x,y,z)$ being the velocity field at the point (x,y,z). Now, when one drops a leaf into the fluid, the vector $\mathrm{curl}(\vec{F})(x,y,z)$ measures the tendency of the leaf to rotate as it moves through the fluid.

In order to help remembering the definition of $\mathrm{curl}(\vec{F})$, the following symbolical notation is used:

$$\boxed{\mathrm{curl}(\vec{F}) = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}} .$$

Example 0.1 Let $\vec{F}(x, y, z) = (y, x, z^2)$. Find Evaluate $\text{curl}(\vec{F})(x, y, z)$ for every $(x, y, z) \in \mathbb{R}^3$,. We find that

$$\text{curl}(\vec{F})(x, y, z) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ y & x & z^2 \end{vmatrix} = (0, 0, 0)$$

Theorem 0.2 *Let U be subset of $\mathbb{R}^3$ and $f : U \to \mathbb{R}$ be with continuous second order partial derivatives. Then*

$$\text{curl}(\nabla f) = \vec{0}\,.$$

Proof:

$$\begin{aligned} \text{curl}(\vec{F}) &= \nabla \times \nabla f = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ \dfrac{\partial f}{\partial x} & \dfrac{\partial f}{\partial y} & \dfrac{\partial f}{\partial z} \end{vmatrix} \\ &= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}, \frac{\partial^2 f}{\partial x \partial z} - \frac{\partial^2 f}{\partial z \partial x}, \frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) = (0, 0, 0)\,. \end{aligned}$$

Remark 0.3 Theorem 0.2 implies that if a vector field $\vec{F} : U \subset \mathbb{R}^3 \to \mathbb{R}^3$ is such that $\text{curl}(\vec{F}) \neq \vec{0}$, then $\vec{F}$ is not conservative. However, if $\text{curl}(\vec{F}) = \vec{0}$ and the subset U is simply connected (i.e., U does not have "holes"), one can prove that the vector field $\vec{F}$ is conservative.

The sets $\mathbb{R}^3$, $\{(x, y, z) \in \mathbb{R}^3\colon\ x^2 + y^2 + z^2 \leq 1\}$, $\{(x, y, z) \in \mathbb{R}^3\colon\ z \geq 0\}$ are simply connected while the ring $\{(x, y, z) \in \mathbb{R}^3\colon\ 1 \leq x^2 + y^2 + z^2 \leq 4\}$ is not.

For a two-dimensional vector field $\vec{F} : D \subset \mathbb{R}^2 \to \mathbb{R}^2$ with $F(x, y) = (P(x, y), Q(x, y))$ we write it as a three-dimmensional vector field and get

$$\vec{F}(x, y) = (P(x, y), Q(x, y), 0)\,.$$

For such a vector field, we obtain

$$\boxed{\text{curl}(\vec{F}) = \left(0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\vec{k}}\,. \tag{2}$$

Using (2), we can then write the vector form of identity of the Green theorem to get

$$\boxed{\oint_C \vec{F} \cdot d\vec{r} = \iint_D \text{curl}(\vec{F}) \cdot \vec{k}\, dx dy}\,. \tag{3}$$

Notice that the identity (3) is just a particular case of a more general identity from called the **Stokes' theorem** which will be presented in the next section.

Let $\vec{F} : U \to \mathbb{R}^3$ be a vector field with

$$\vec{F}(x, y, z) = (P(x, y, z), Q(x, y, z), R(x, y, z)) \quad \forall (x, y, z) \in U$$

with the functions P, Q, R having continuous partial derivatives. **The divergence** of the vector field $\vec{F}$ is the scalar field defined by

$$\boxed{\operatorname{div}(\vec{F}) = \underbrace{\nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \cdot (P, Q, R)}_{\text{symbolical}} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}}\,. \tag{4}$$

In Fluid dynamic, the divergence of velocity field of the fluid measure the variation of the flux of the fluid.

Exercise 0.4 Let $F = (P, Q, R) : \mathbb{R}^3 \to \mathbb{R}^3$ be a vector field with the components P, Q, R having continuous second order partial derivatives. Show that

$$\boxed{\operatorname{div}\big(\operatorname{curl}(\vec{F})\big) = 0}\,.$$

0.2.2 Orientation of a surface

Below are examples of surfaces classified in two categories:

Piecewise regular surfaces

We recall that a surface $\mathcal{S}$ is called **piecewise regular** if $\mathcal{S}$ is finite union of regular surfaces. For instance, a rectangular box is piecewise regular as each face is a regular surface and there is no normal vector at any point of the edges, and the normal vector jumps of 90 degrees as we move from any face to an adjacent face.

Orientation of a system of coordinates

Consider the system of coordinates x, y, z for which there are only two possible orientations according to the right-hand rule:

(1) We place the right-hand in such a way that the thumb points up along the z-axis and the remaining fingers rotate from the positive x-axis to the positive y-axis.

(2) In the other orientation, the positive z-axis is downward.

Any of these two orientations of the coordinate system x, y, z can be chosen as the positive orientation of the system while the other will be called negative. However, the orientation (1) is quite often considered as positive.

The orientation of a regular surface

Now, let $\mathcal{S}$ be the regular surface described by a vector function $\vec{r} : D \subset \mathbb{R}^2 \to \mathbb{R}^3$. We recall that, this means that:

$$\begin{cases} \text{both } \vec{r}_s \text{ and } \vec{r}_t \text{ are continuous and that} \\ \vec{r}_s(s, t) \times \vec{r}_t(s, t) \neq (0, 0, 0) \quad \forall (s, t) \text{ in the interior of the domain } D\,. \end{cases}$$

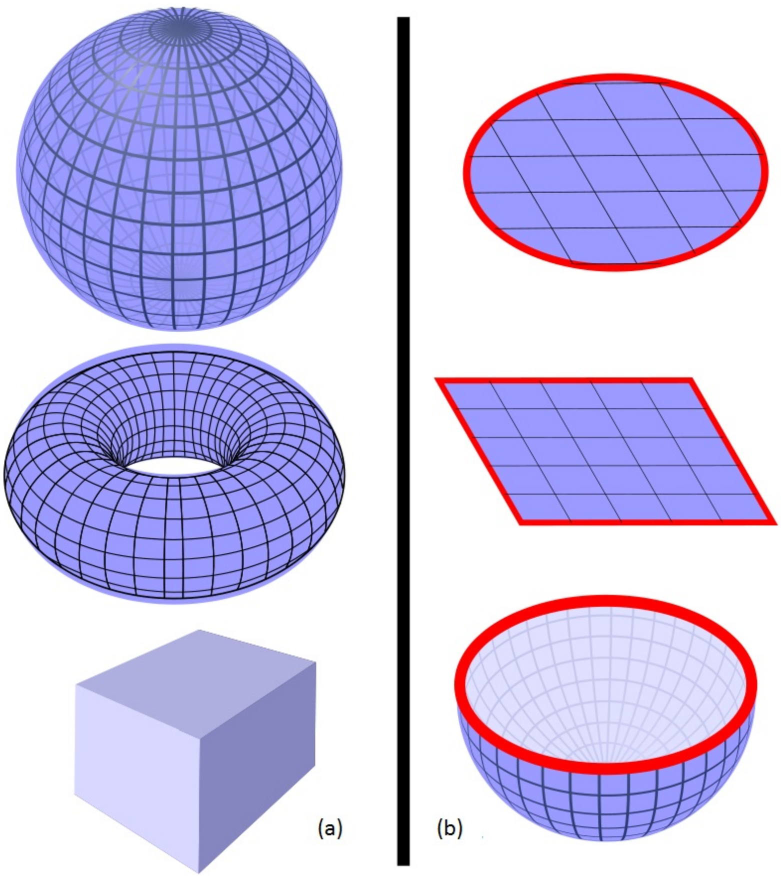

Figure 1: (a) Examples of Surfaces $\mathcal{S}$ without boundary: closed surfaces; (b) Examples of Surfaces $\mathcal{S}$ with boundary $\mathcal{C} = \partial\mathcal{S}$ shown in red. The picture is extracted from https://en.wikipedia.org/wiki/Surface-(topology).

So, at every point (x_0, y_0, z_0) on the regular surface $\mathcal{S}$ with $(x_0, y_0, z_0) = \vec{r}(s,t)$ for some $(s,t) \in D$ are associated three vectors

$$\boxed{\vec{r}_s(s,t), \quad \vec{r}_t(s,t), \quad \vec{r}_s(s,t) \times \vec{r}_t(s,t)}.$$

Definition 0.5 We say that the regular surface S is orientable if the normal vector $\vec{r}_s(s,t) \times \vec{r}_t(s,t)$ points continuously to the same side of the surface, as one moves on the surface.

There is one popular of example of a non-orientable surface called the **Möbius strip**, which can be created by taking a paper strip, giving one end a half-twist, and then joining the ends to form a loop.

For an orientable surface S, the positive orientation is given according to the vectors $\vec{r}_s(s,t), \vec{r}_t(s,t)$ and $\vec{r}_s(s,t) \times \vec{r}_t(s,t)$ according to right-hand rule with the thumb pointing in the direction of the normal $\vec{r}_s(s,t) \times \vec{r}_t(s,t)$ and the other fingers rotated from $\vec{r}_s(s,t)$ to $\vec{r}_t(s,t)$.

Notice that if $\mathcal{S}$ is a surface with boundary, the positive orientation of such a surface corresponds to a counterclockwise orientation of its boundary.

When the surface $\mathcal{S}$ is oriented, its normal vector points to the side called the **positive side** of the surface $\mathcal{S}$ denoted by $\mathcal{S}^+$ while the other side of the surface is called **its negative side** and is denoted by $\mathcal{S}^-$.

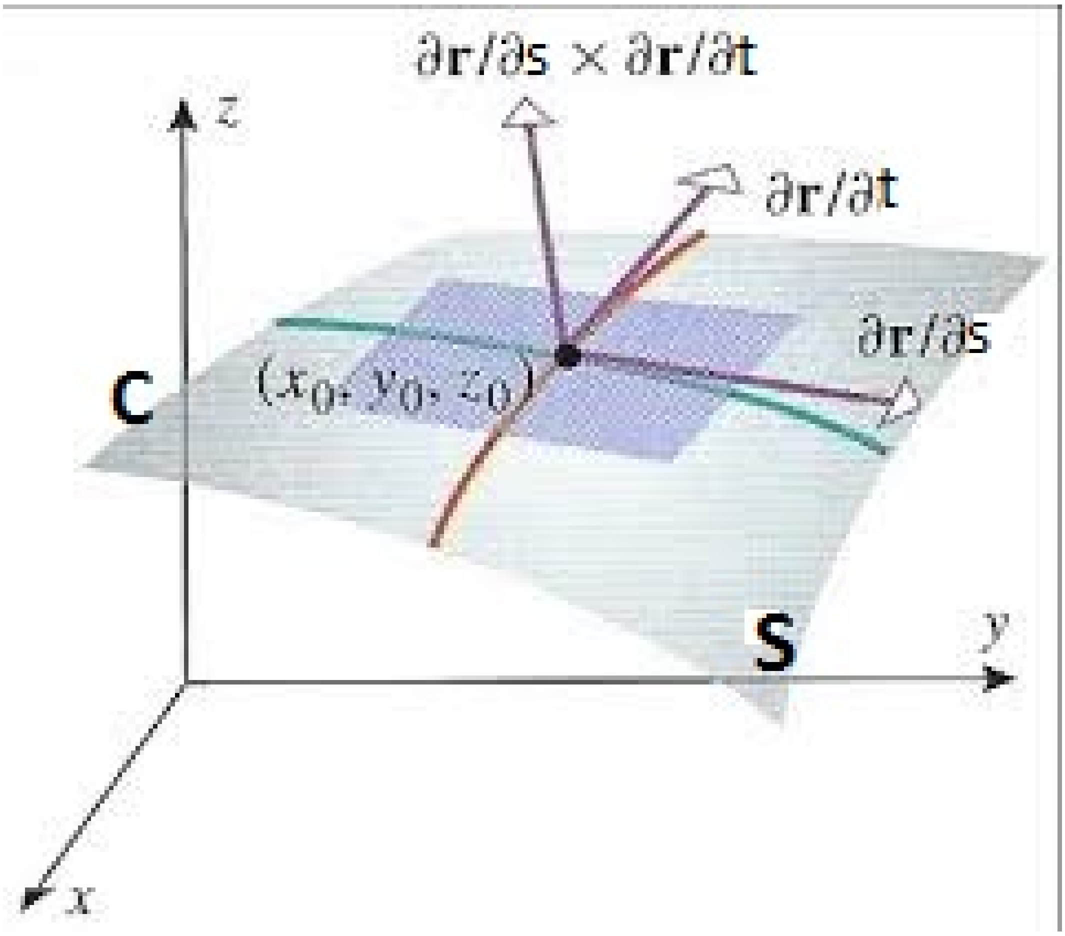

Figure 2: Oriented surface **S** with normal $\vec{r}_s(s,t)\times\vec{r}_t(s,t)$ at the point $(x_0,y_0,z_0)=\vec{r}(s,t)$ and boundary **C**. The original picture is extracted from https://www.maths.tcd.ie/ frolovs/Calculus/2E02-Multiple-Integrals-III.pdf and adapted to our notations.

Examples 0.6

(a) Let Σ be the surface described by the vector function

$$\vec{r}(s,t) := (\cos s\sin t, \sin s\sin t, \cos t) \quad \text{for } 0\le s\le \pi \text{ and } 0\le t\le\pi\,.$$

Then

$$\Sigma = \big\{(x,y,z)\in\mathbb{R}^3\colon\; x^2+y^2+z^2=1,\;\; y\ge 0\big\}\,.$$

$$\vec{r}_s(s,t) = (-\sin s\sin t, \cos s\sin t, 0), \quad \vec{r}_t(s,t) = (\cos s\cos t, \sin s\cos t, -\sin t)$$

$$\begin{aligned}
\vec{r}_s(s,t)\times\vec{r}_t(s,t) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\sin s\sin t & \cos s\sin t & 0 \\ \cos s\cos t & \sin s\cos t & -\sin t \end{vmatrix} \\
&= (-\cos s\sin^2 t, -\sin s\sin^2 t, -\sin^2 s\sin t\cos t - \cos^2 s\sin t\cos t) \\
&= (-\cos s\sin^2 t, -\sin s\sin^2 t, -\sin t\cos t) \\
&= -\sin t(\cos s\sin t, \sin s\sin t, \cos t) = -\sin t\,\vec{r}(s,t)
\end{aligned}$$

Since $-\sin t\le 0$, then the normal vector $\vec{r}_s(s,t)\times\vec{r}_t(s,t)$ opposite to $\vec{r}(s,t)$ and therefore $\vec{r}_s(s,t)\times\vec{r}_t(s,t)$ points towards the inside of the sphere. Therefore, the inside of the sphere is the positive side of the surface Σ.

(b) Let $\mathcal{S}$ be the graph of the function $f(x,y) = x^2 + y^2$ defined on the disc $D := \{(x,y) \in \mathbb{R}^2 \colon x^2 + y^2 \leq 1\}$. Then $\mathcal{S}$ is defined by the vector function

$$\vec{r}(x,y) = (x, y, x^2 + y^2) \quad \text{for } (x,y) \in D$$

for which

$$\vec{r}_x(x,y) = (1, 0, 2x), \quad \vec{r}_y(x,y) = (0, 1, 2y)$$

and

$$\vec{r}_x(x,y) \times \vec{r}_y(x,y) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2x \\ 0 & 1 & 2y \end{vmatrix} = (-2x, -2y, 1)\,.$$

Also, in this case we have an inner normal. In fact, $\vec{r}_s(0,0) \times \vec{r}_t(0,0) = (0,0,1)$; so, the z-axis is normal to $\mathcal{S}$ at the point $(0,0,0)$. So the inside of the half-sphere $\mathcal{S}$ is the positive side of the surface.

0.2.3 Surface integrals

In this section, we want to extend the notion of double integrals over surfaces in the space which do not lie entirely on the xy-plane. Precisely, let $\mathcal{S}$ be for instance a regular surface described by the vector function $\vec{r} : D \subset \mathbb{R}^2 \to \mathbb{R}^3$. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a continuous function and let $\vec{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be a continuous vector field. We would like to define the integrals

$$\boxed{\iint_{\mathcal{S}} f(x,y,z) dS \quad \text{and} \quad \iint_{\mathcal{S}} \vec{F}(x,y,z) \cdot \vec{n}(x,y,z)\, dS},$$

where $\vec{n}(x,y,z)$ denotes the unit normal vector to the surface $\mathcal{S}$ at the point $(x,y,z) \in \mathcal{S}$. We recall that the normal vector to the surface $\mathcal{S}$ at the point $(x,y,z) = \vec{r}(s,t)$ is given by

$$\vec{r}_s(s,t) \times \vec{r}_t(s,t)$$

and therefore, the unit normal vector at that point is:

$$\boxed{\vec{n}(x,y,z) := \frac{\vec{r}_s(s,t) \times \vec{r}_t(s,t)}{\|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\|} \quad \text{for} \quad (x,y,z) = \vec{r}(s,t)}\,. \tag{5}$$

By analogy with the construction of the double integral over a domain on the xy-plane, as studied in the previous chapter, let us first consider the problem of finding the area of the surface $\mathcal{S}$.

To this aim, we assume that the domain $D = [a,b] \times [c,d]$ and consider a partition

$$a = s_0 < s_1 < s_2 < \ldots < s_{n-1} < s_n = b \quad \text{of} \quad [a,b] \;\; \text{with} \;\; \Delta s_i = s_i - s_{i-1}\,,$$

and

$$c = t_0 < t_1 < t_2 < \ldots < t_{m-1} < t_m = c \quad \text{of} \quad [c,d] \;\; \text{with} \;\; \Delta t_j = t_j - t_{j-1}\,.$$

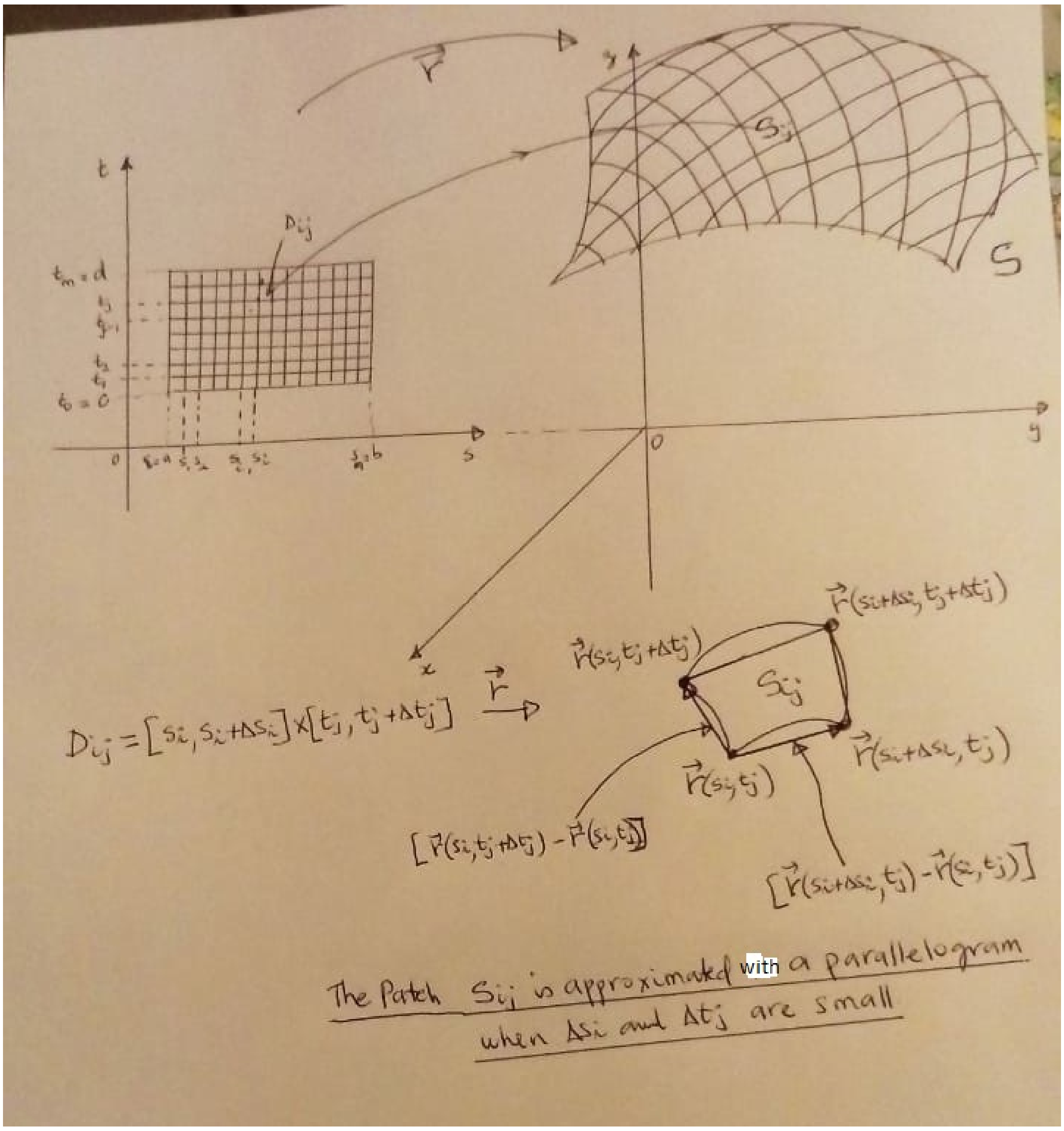

Figure 3: A constructive approximation of the area of the surface $\mathcal{S}$

Hence, we get a grid of the rectangle D into mn subrectangles D_{ij}. To this grid of the rectangle D, there corresponds a partition of the surface $\mathcal{S}$ into a finite number of patches $\mathcal{S}_{ij} = \vec{r}(D_{ij})$ (see Figure 3). We obtain that

$$A(\mathcal{S}) = \sum_{ij} A(\mathcal{S}_{ij})\,.$$

Now, as m, n are becoming large, the patches $\mathcal{S}_{ij}$ are becoming smaller and we write approximate the quantity $A(\mathcal{S}_{ij})$ with the area of the paralellogram with vertices

$$\vec{r}(s_i, t_j), \qquad \vec{r}(s_i+\Delta s_i, t_j), \qquad \vec{r}(s_i, t_j+\Delta t_j), \qquad \vec{r}(s_i+\Delta s_i, t_j+\Delta t_j)\,.$$

Using the formula of the area of a parallelogram, we get

$$\begin{aligned} A(\mathcal{S}_{ij}) &\approx \left\| \left[\vec{r}(s_i+\Delta s_i, t_j) - \vec{r}(s_i, t_j)\right] \times \left[\vec{r}(s_i, t_j+\Delta t_j) - \vec{r}(s_i, t_j)\right] \right\| \\ &\approx \left\| \left[\vec{r}_s(s_i^*, t_j^*)\Delta s_i\right] \times \left[\vec{r}_t(s_i^*, t_j^*)\Delta t_j\right] \right\| = \|\vec{r}_s(s_i^*, t_j^*) \times \vec{r}_t(s_i^*, t_j^*)\| \Delta s_i \Delta t_j\,. \end{aligned}$$

Therefore, we obtain the following approximation of the area of $\mathcal{S}$ and n, m are set large

$$\boxed{A(\mathcal{S}) \approx \sum_{i=1}^{n}\sum_{j=1}^{m} \|\vec{r}_s(s_i^*, t_j^*) \times \vec{r}_t(s_i^*, t_j^*)\| \Delta s_i \Delta t_j}$$

the later is a Riemann sum of the function

$$\boxed{(s,t) \in D \quad \longrightarrow \quad \|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\|}.$$

Therefore, we finally obtain the following formula for the area of the surface $\mathcal{S}$

$$\boxed{A(\mathcal{S}) = \iint_D \|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\|\, dsdt}. \tag{6}$$

We can extend that process to the evaluation of the surface integrals

$$\boxed{\iint_{\mathcal{S}} f(x,y,z) dS = \iint_D f(\vec{r}(s,t)) \|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\|\, dsdt} \tag{7}$$

and

$$\boxed{\iint_{\mathcal{S}} \vec{F}(x,y,z) \cdot \vec{n}(x,y,z) dS = \iint_D \vec{F}(\vec{r}(s,t)) \cdot \frac{\vec{r}_s(s,t) \times \vec{r}_t(s,t)}{\|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\|} \|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\|\, dsdt}$$

Hence,

$$\boxed{\iint_{\mathcal{S}} \vec{F}(x,y,z) \cdot \vec{n}(x,y,z) dS = \iint_D \vec{F}(\vec{r}(s,t)) \cdot \left[\vec{r}_s(s,t) \times \vec{r}_t(s,t)\right] dsdt} \tag{8}$$

Definition 0.7 If $\mathcal{S}$ is a piecewise regular surface, then

$$\iint_{\mathcal{S}} f(x,y,z)\, dS = \sum_{i=1}^{n} \iint_{\mathcal{S}_i} f(x,y,z)\, dS \tag{9}$$

$$\iint_{\mathcal{S}} \vec{F}(x,y,z) \cdot \vec{n}(x,y,z)\, dS = \sum_{i=1}^{n} \iint_{\mathcal{S}_i} \vec{F}(x,y,z) \cdot \vec{n}(x,y,z)\, dS \tag{10}$$

Remark 0.8

1. Whenever the surface $\mathcal{S}$ is the graph of a function $g : D \subset \mathbb{R}^2 \to \mathbb{R}$ with continuous first order partial derivatives, then recall (from the lectures on parameric surfaces) that $\mathcal{S}$ is described by the vector function

$$\vec{r} : D \subset \mathbb{R}^3 \to \mathbb{R}^3$$

   defined by

$$\vec{r}(x,y) := (x, y, g(x,y)) \qquad \forall (x,y) \in D$$

   and that

$$\left.\begin{array}{l} \vec{r}_x(x,y) = (1, 0, g_x(x,y)), \\ \vec{r}_y(x,y) := (0, 1, g_y(x,y)) \end{array}\right\} \Rightarrow \boxed{\vec{r}_x(x,y) \times r_y(x,y)) = (-g_x(x,y), -g_y(x,y), 1)}$$

$$\Rightarrow \boxed{\|\vec{r}_x(x,y) \times r_y(x,y))\| = \sqrt{g_x^2(x,y) + g_y^2(x,y) + 1}}$$

Hence,

$$\iint_{\mathcal{S}} f(x,y,z)\,dS = \iint_D f(x,y,g(x,y))\sqrt{g_x^2(x,y)+g_y^2(x,y)+1}\,dxdy \quad (11)$$

$$\iint_{\mathcal{S}} \vec{F}(x,y,z)\cdot\vec{n}(x,y,z)\,dS = \iint_D \vec{F}(x,y,g(x,y))\cdot(-g_x(x,y),-g_y(x,y),1)dxdy\,. \quad (12)$$

2. Assume that the surface $\mathcal{S}$ is a level surface a function $g : U \subset \mathbb{R}^3 \to \mathbb{R}$ with continuous first order partial derivatives with $\nabla g(x,y,z) \neq (0,0,0)$ for every $(x,y,z) \in U$. Such a surface $\mathcal{S}$ is regular with unit normal vector

$$\boxed{\vec{n}(x,y,z) := \frac{\nabla g(x,y,z)}{\|\nabla g(x,y,z)\|} \qquad \forall (x,y,z) \in U}\,.$$

Definition 0.9 The surface integral $\displaystyle\int_{\mathcal{S}} \vec{F}\cdot\vec{n}\,dS$ is called the flux integral of the vector field $\vec{F}$ across the surface $\mathcal{S}$.

Let us recall our major goal in this section which is extend the vector form of the Green identity in (3), that is,

$$\boxed{\oint_C \vec{F}\cdot d\vec{r} = \iint_D \mathrm{Curl}(\vec{F})\cdot\vec{k}\,dxdy}\,, \quad (13)$$

to more general $3D$-dimensional vector fields with continuous partial derivatives and more general regular surfaces which do not necessarily lye entirely on the xy-plane. This is the purpose of the Stokes' theorem below. Notice that, the right-hand side of (13) is a flux integral.

Theorem 0.10 (The Stokes Theorem)
Let $\mathcal{S}$ be a piecewise smooth, oriented surface, whose oriented boundary $\mathcal{C}$ is a piecewise smooth, simple closed curve. If $\vec{F} = (P,Q,R)$ is a smooth, three-dimensional vector field whose domain is a region in $\mathbb{R}^3$ that contains the surface $\mathcal{S}$, then

$$\boxed{\oint_{\mathcal{C}} \vec{F}(x,y,z)\cdot d\vec{r} = \iint_{\mathcal{S}} \mathrm{curl}(\vec{F})(x,y,z)\cdot\vec{n}(x,y,z)\,dS}\,. \quad (14)$$

As in the case of Green's Theorem, the evaluation of a line integral can be simplified by applying Stokes' Theorem.

Consider for instance, the vector field $\vec{F}(x,y,z) = (-3y, 2z, -x)$ and let us evaluate $\displaystyle\oint_{\mathcal{C}} \vec{F}\cdot d\vec{r}$ where $\mathcal{C}$ is the intersection the plane $z = y-1$ with the cylinder $x^2+y^2 = 4$.
If one has to evaluate the line integral directly, a possible parametrization of $\mathcal{C}$ would be $x = 2\cos t$, $y = 2\sin t$ and $z = 2\sin t - 1$, with $0 \le t \le 2\pi$. The direct evaluation of the line integral would have been too long **(Check it!)**.

But in order to apply Stokes' theorem, let $\mathcal{S}$ be the part of the surface $z = y-1$ that is enclosed by the curve $\mathcal{C}$ and let D be the projection of $\mathcal{S}$ onto the xy-plane.

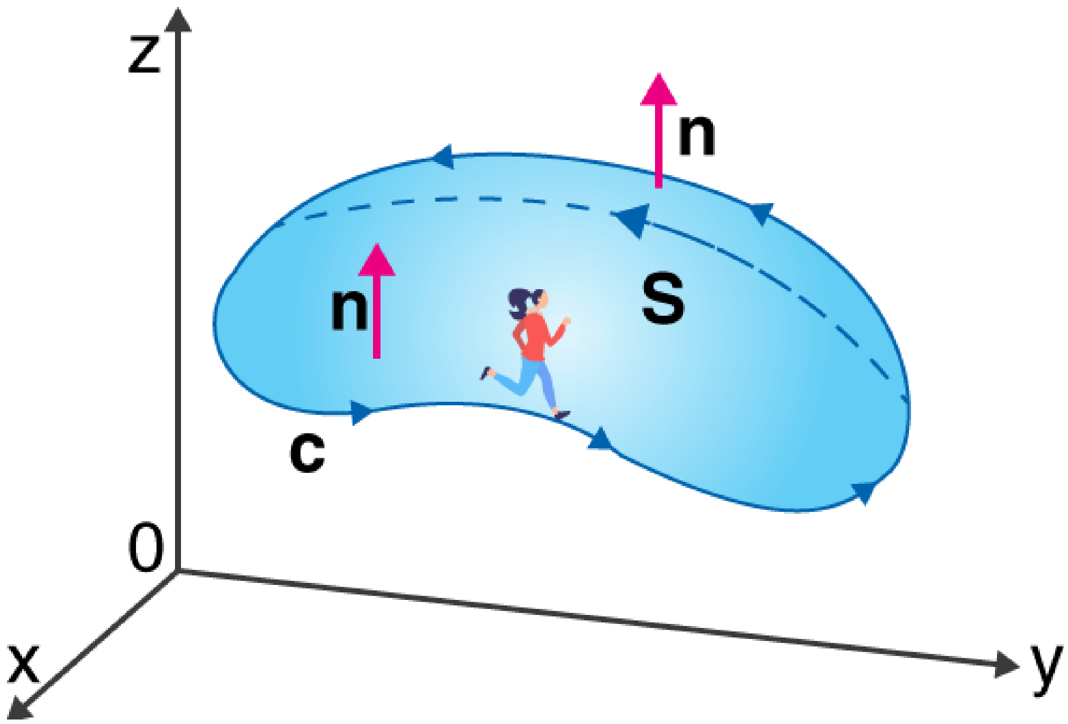

Figure 4: The oriented surface **S** with boundary **C**, to which the stokes theorem is applied. The picture is extracted from https://byjus.com/maths/stokes-theorem/.

$\operatorname{curl}\vec{F} = (-2,1,3)$, and the normal to $\mathcal{S}$ is given by $\vec{n} = \frac{1}{\sqrt{2}}(0,-1,1)$. Hence, by Stokes' theorem we get

$$\begin{aligned}\oint_{\mathcal{C}} \vec{F}\cdot d\vec{r} &= \iint_{\mathcal{S}} \operatorname{curl}\vec{F}\cdot\vec{n}\,dS = \iint_{\mathcal{S}} \frac{1}{\sqrt{2}}(-2,1,3)\cdot(0,-1,1)\,dS \\ &= \sqrt{2}\iint_{\mathcal{S}} dS = \sqrt{2}\iint_{D}\sqrt{2}dxdy = 8\pi\,.\end{aligned}$$

Exercise 0.11 Use Stokes' Theorem to evaluate $\oint_{\mathcal{C}} z^2 e^{x^2}dx + xy^2dy + \arctan y\,dz$ where $\mathcal{C}$ is the circle $x^2+y^2=9$, by finding the surface $\mathcal{S}$ with $\mathcal{C}$ as its boundary **(Take the simplest surface)** and such that the orientation of $\mathcal{C}$ is counterclockwise as viewed from above. **Answer:** $\boxed{\dfrac{81\pi}{4}}$. **Check the answer by evaluating also directly the line integral!)**

The next set of notes will be on the divergence theorem!